

Lecture 5: Air Pollution

ECON7720: Ecological and Environmental Economics

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Semester 2, 2026 | Required reading: Tietenberg & Lewis, Ch. 8

Recap of Lectures 3–4

- The **efficient** pollution level: $MD(E^*) = MAC(E^*)$.
- **Cost-effectiveness**: the equimarginal principle ($MAC_i = t$ or $MAC_i = p$ under tax/cap).
- **Weitzman**: steep MD $\Rightarrow$ quantity; steep MAC $\Rightarrow$ price; collars blend the two.
- **Location**: when it matters, equalise MAC_i/a_i (ambient charges/permits, trading ratios, hotspots).
- **Pollutant taxonomy**: stock vs fund, uniform vs non-uniform mixing $\Rightarrow$ guides instrument design.
- **Coase**: private bargaining reaches E^* regardless of the property right, but transaction costs $\Rightarrow$ government instruments.
- **Instruments**: CAC (standards), emission taxes, abatement subsidies, cap-and-trade, each with different efficiency, cost-effectiveness, and distributional properties.

Lecture Outline

1. **Motivation and context:** conventional pollutants, $PM_{2.5}$, global exposure, state capacity, and why it matters for development.
2. **Command-and-control approaches:** NAAQS, standards across jurisdictions, efficiency and cost-effectiveness.
3. **Market-based approaches:** emission charges, trading mechanisms, the US Acid Rain Program, co-benefits, and developing-country applications.

Learning Resources

Required readings

- Tietenberg & Lewis, Ch. 8 (*Stationary-Source Local and Regional Air Pollution*): read and attempt the self-test exercises.

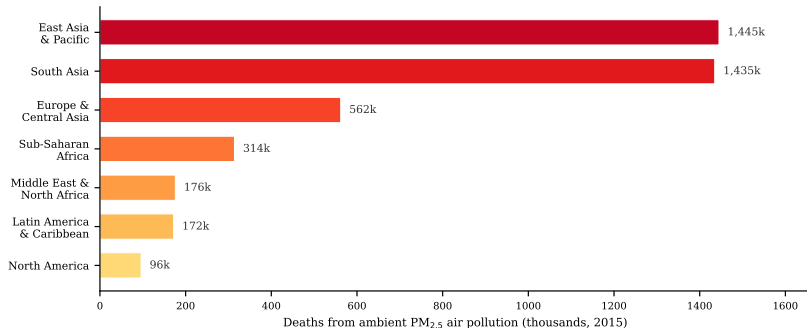
Recommended readings

- Schmalensee, R. & Stavins, R.N. (2013), "The SO₂ Allowance Trading System: The Ironic History of a Grand Policy Experiment," *Journal of Economic Perspectives* 27(1), 103–122.

Motivation and Context

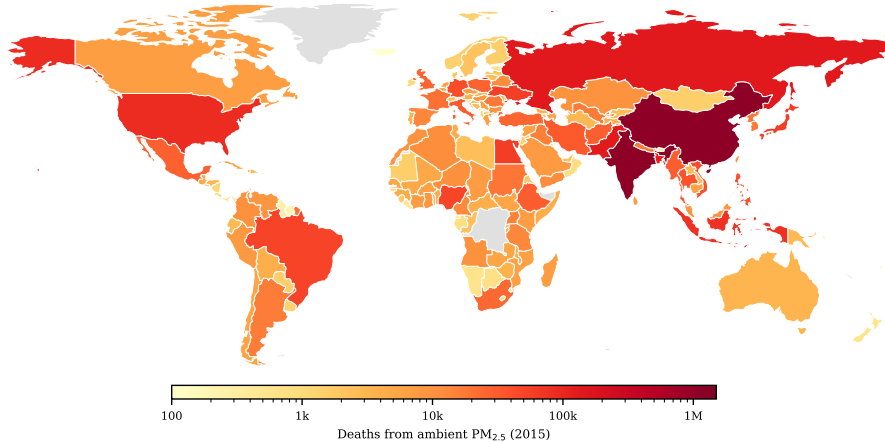
Why air pollution matters (1/3)

- The WHO estimates outdoor (ambient) air pollution causes ~ 4.7 million premature deaths per year worldwide (GBD 2021; the widely cited 8.1 million figure includes household air pollution).
- Ozone and fine particulate matter ($PM_{2.5}$) are the most important pollutants for human health.
- The damages are not limited to respiratory illness: evidence links air pollution to cognitive decline, productivity loss, and crime.



Source: OWID / IHME State of Global Air, GBD 2015 country-level estimates.

Why air pollution matters (2/3)



China (1.1M) and **India** (1.1M) together account for over half of global deaths from ambient PM_{2.5}. Large absolute counts also in Russia, Pakistan, Bangladesh, and Indonesia, driven by population size, high exposure, and limited abatement.

Source: OWID / IHME State of Global Air, GBD 2015 country-level estimates. Log scale.

Why air pollution matters (3/3)

India as a case study

- More than 90% of India's population lives in areas where air quality is below WHO standards.
- 18% of deaths in India in 2019 were attributable to air pollution (ICMR/*Lancet Planetary Health*).
- Air pollution shortens average life expectancy in India by ~6 years, and up to ~10 years in Delhi (AQLI, relative to WHO standards).



Source: Wikimedia Commons, CC BY-SA 4.0. Delhi smog, November 2019.

India accounts for the largest share of global air pollution deaths, over 1 million per year, driven by rapid industrialisation, urbanisation, and biomass burning.

Criteria (conventional) pollutants

The six regulated substances (US)

Pollutant	Key concern	Type
Sulfur dioxide (SO ₂)	Acid rain, respiratory	Local/regional
PM ₁₀ / PM _{2.5}	Cardiovascular, premature death	Local
Carbon monoxide (CO)	Acute poisoning	Local
Ground-level ozone (O ₃)	Smog, respiratory	Regional
Nitrogen dioxide (NO ₂)	Smog precursor, respiratory	Local/regional
Lead (Pb)	Neurotoxic, child development	Local

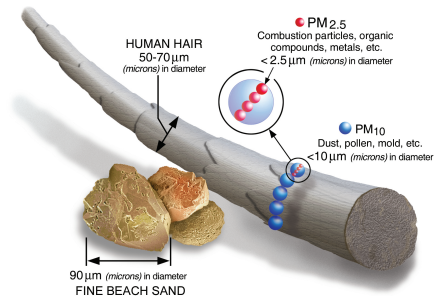
These are called **criteria** pollutants because the regulator must publish “criteria documents” reviewing the science before setting standards. They are dangerous mainly at **high concentrations**, classic **non-uniformly mixed**, local/regional pollutants.

PM_{2.5} — the most dangerous pollutant

What is particulate matter?

- PM₁₀: particles $\leq 10\ \mu\text{m}$ (dust, pollen). Penetrates the upper respiratory tract.
- PM_{2.5}: particles $\leq 2.5\ \mu\text{m}$ (combustion products, secondary aerosols). Penetrates deep into the lungs and enters the bloodstream.
- PM_{2.5} is the primary indicator used by IQAir and WHO guidelines.
- Sources: coal/oil burning, vehicles, dust storms, wildfires, and **secondary formation** from SO₂/NO_x.

Controlling SO₂ and NO_x reduces PM_{2.5}, a major **co-benefit** of acid rain policy.



Source: US EPA (public domain).

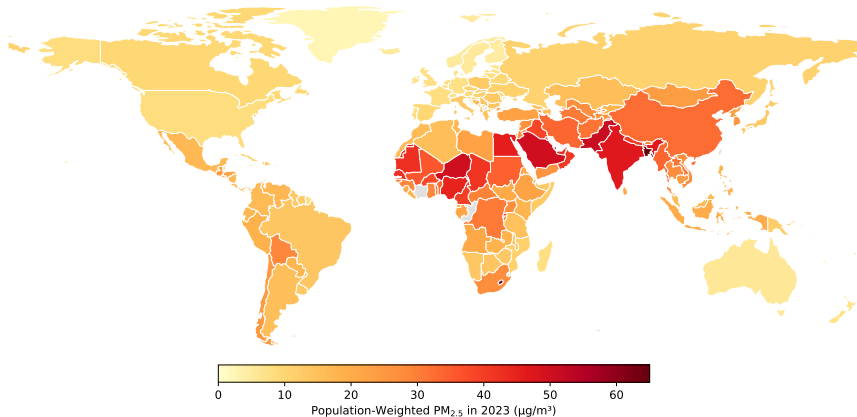
IQAir 2024 World Air Quality Report

Global PM_{2.5} concentrations

- The report measures annual average PM_{2.5} across cities with publicly available monitoring data.
- **WHO guideline:** annual average PM_{2.5} $\leq 5 \mu\text{g}/\text{m}^3$.
- Most polluted countries (annual avg): **Chad** (~ 92), **Bangladesh** (~ 78), **Pakistan** (~ 74), **India** (~ 51) $\mu\text{g}/\text{m}^3$ — 10–18 $\times$ the WHO guideline.
- **Australia:** met the WHO guideline in 2024, but wildfire episodes cause severe spikes.
- Only 7 countries met the WHO guideline in 2024 (Australia, Bahamas, Barbados, Estonia, Grenada, Iceland, New Zealand); 91% exceeded it.

Source: [IQAir](#) is a Swiss air quality monitoring platform that aggregates publicly available ground-station data into annual country and city rankings.

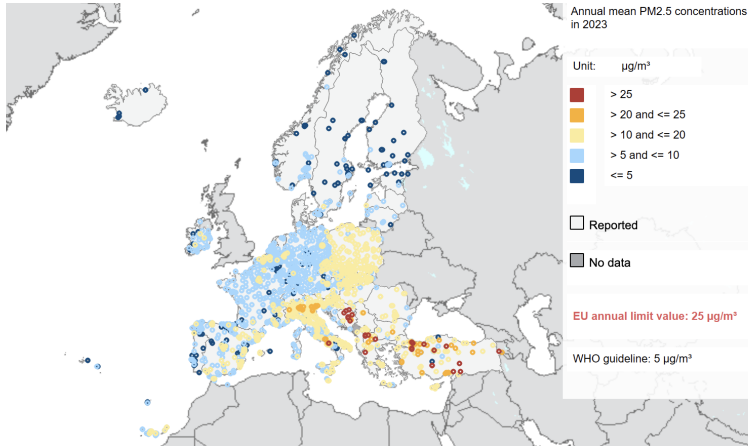
Global PM_{2.5} concentrations (2023)



Population-weighted PM_{2.5} in 2023: Pakistan 51 · India 47 · Egypt 46 · China 33 · Indonesia 20 vs. WHO guideline 5 µg/m³.

"Population-weighted" = each grid cell weighted by population, so the average reflects what the *typical person* breathes, not the geographic mean. Satellite-derived estimates (Shen et al. 2024, *ACS ES&T Air*). Data: ACAG, Washington University in St. Louis (CC BY 4.0).

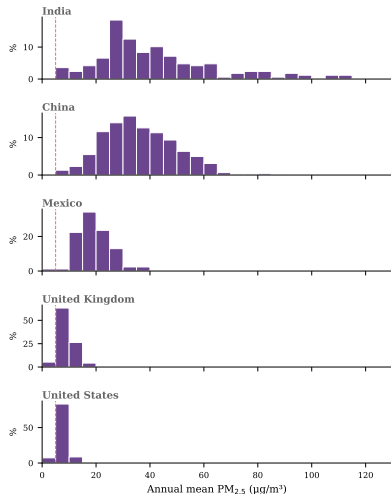
PM_{2.5} concentrations in Europe (2023)



Station-level annual mean PM_{2.5}. Most of western and northern Europe meets the EU limit ($25 \mu\text{g}/\text{m}^3$) but **not** the WHO guideline ($5 \mu\text{g}/\text{m}^3$). The Po Valley, Balkans, and Türkiye remain hotspots, driven by solid-fuel heating and older vehicle fleets.

Source: [European Environment Agency](#) (CC BY 1.0), 2023 data.

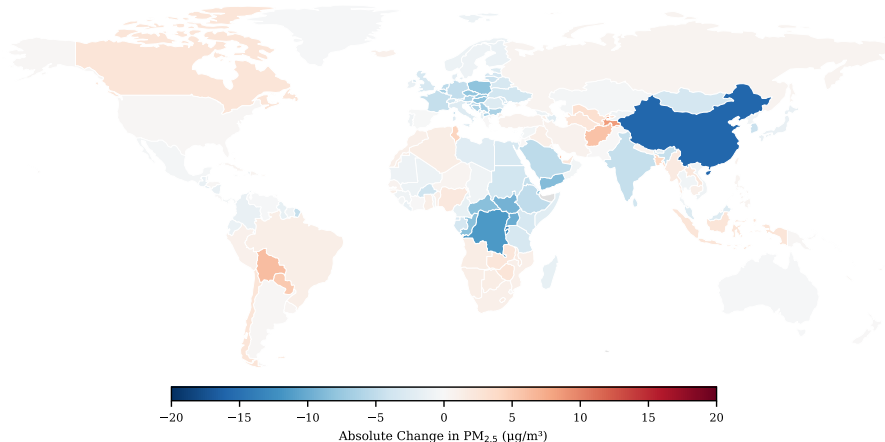
City-level PM_{2.5} distributions



- City-level PM_{2.5} (ground monitors): India mean 42, China 37, Mexico 19, UK 9, US 8 µg/m³.
- Cities out of compliance with the WHO guideline exist in every country shown.
- But the distributions barely overlap: India's cleanest cities are dirtier than America's dirtiest.
- This is not a marginal difference, it requires structural transformation of energy, transport, and industry.

Source: WHO Ambient Air Quality Database V8.0 (June 2026). Most recent annual mean per city. Red dashed line: WHO guideline (5 µg/m³).

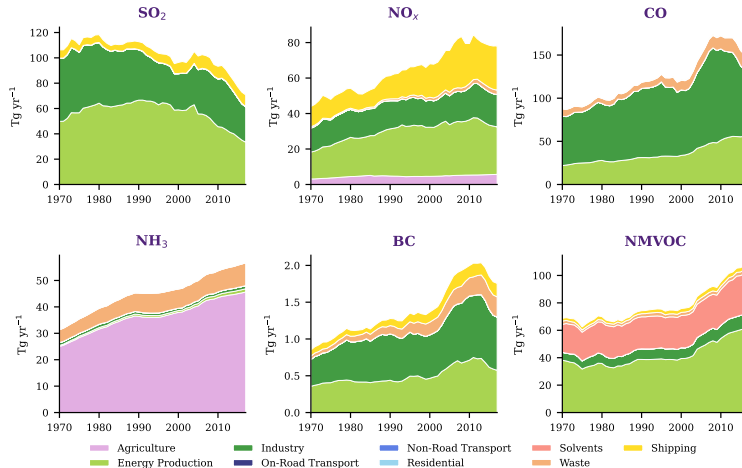
Change in PM_{2.5} concentrations (2013–2023)



China ($-16 \mu\text{g}/\text{m}^3$) achieved the largest absolute improvement, driven by aggressive coal controls and industrial restructuring. Sub-Saharan Africa and parts of South America **worsened**. Most developed countries show modest declines.

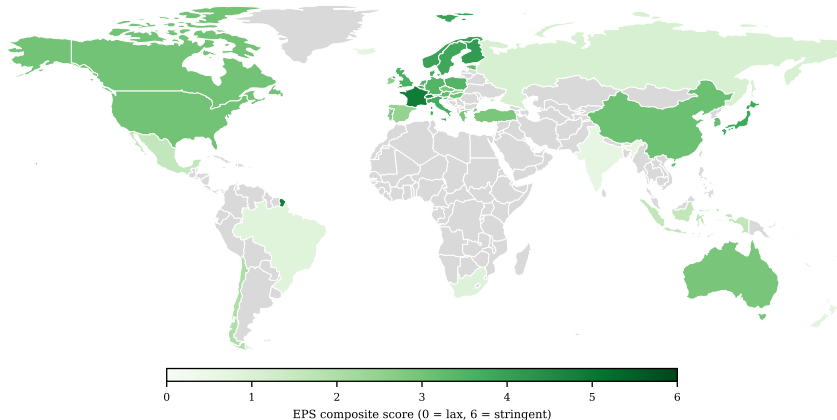
Source: OWID / IHME State of Global Air. Population-weighted satellite-derived PM_{2.5} estimates.

Global air pollutant emissions by sector (1970–2017)



Source: CEDS GBD-MAPS v1 (McDuffie et al., 2020, *Earth System Science Data*). Global anthropogenic emissions by sector in Tg yr^{-1} (million tonnes per year). SO_2 = sulfur dioxide; NO_x = nitrogen oxides; CO = carbon monoxide; NH_3 = ammonia; BC = black carbon; NMVOC = non-methane volatile organic compounds.

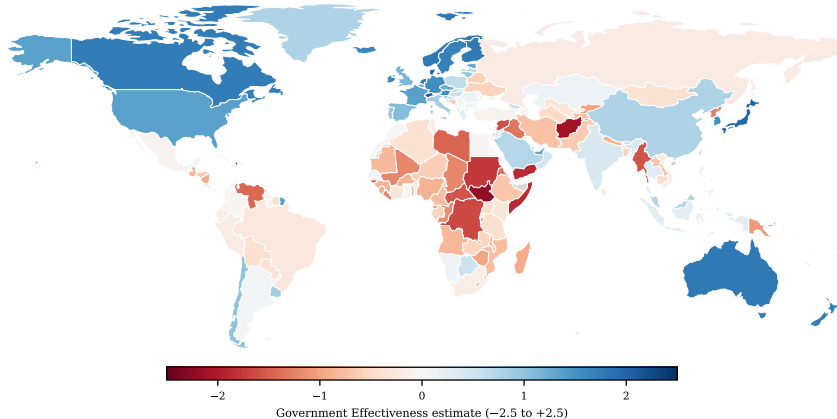
Environmental policy stringency across countries



The OECD Environmental Policy Stringency index measures market-based instruments (taxes, trading schemes) and non-market instruments (standards, R&D subsidies). Europe and the Nordics score highest; most developing countries lack data.

Source: OECD Environmental Policy Stringency Index (Kruse et al., 2022). Composite score: 0 (lax) to 6 (stringent). Data for 40 OECD/partner countries, most recent year available (2018–2020).

Government effectiveness across countries



The countries with the **worst air quality** (South Asia, Sub-Saharan Africa) are the same countries with the **lowest government effectiveness**. Designing environmental policy in these contexts requires instruments that are **simple to administer** and robust to weak enforcement capacity.

Source: World Bank Worldwide Governance Indicators (Kaufmann & Kraay, 2023). Government Effectiveness captures perceptions of public service quality, policy implementation, and civil service independence.

The global picture: air pollution in developing countries

Where the burden is heaviest

- While developed-country air quality has improved dramatically, many cities in South and East Asia now face **severe** PM_{2.5} and ozone levels.
- Cost-effective instruments, emission charges, tradeable permits, are **even more valuable** where budgets are tight and abatement-cost heterogeneity is large.



Source: Wikimedia Commons (Bobak), CC BY-SA 2.5. Same location in Beijing, August 2005: after rain (left) vs. smog (right).

Within countries, exposure tracks poverty

Environmental justice and air pollution

The insulated

- Purifier at home → air-conditioned car → filtered office.
- Wealthier households can **buy** their way out of exposure.

The exposed

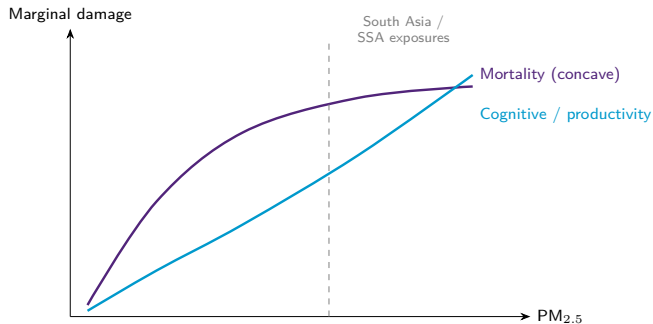
- Low-quality housing, outdoor work (construction, street markets).
- **716 million** people living on <\$1.90/day reside in areas with unsafe PM_{2.5}.

Within-country gaps in exposure are often **larger** than between-country gaps. Air pollution is both an environmental *and* a **distributional** problem.



Source: Wikimedia Commons, GODL-India. Motorcyclist in Delhi smog, January 2006.

The dose–response curve differs by outcome



Mortality flattens at high exposure (frailty selection, saturation, avoidance). **Cognitive and productivity** stay linear or accelerate. Where mortality plateaus, **non-health damages dominate**, the usual damage estimates understate the true cost.

Miller et al. (2025); Zivin & Neidell (2012); Adhvaryu et al. (2022); Zhang et al. (2018).

What weak enforcement looks like in practice

Three failures of state capacity

Corrupted information

- Third-party auditors, paid by the firms they audit, **systematically under-report** (Duflo et al. 2013).
- Beijing air data **falsified** until 2012, “Blue Sky Days” targets (Stoerk 2016).
- Bribed vehicle emissions tests in Mexico City (Oliva 2015).

Missing staff

- Indian pollution control boards: severe shortfalls in sanctioned posts.
- Higher population density $\Rightarrow$ **more people per inspector**.
- India’s clean-air budget (NCAP): USD 90 million, a fraction of the US EPA.

Enforcement gap

- Pass/fail compliance with harsh penalties (e.g. plant shutdown) $\Rightarrow$ regulators hesitate to enforce, so penalties are **rarely applied**.
- The gap between law on paper and enforcement on the ground is a **binding constraint**.

Command-and-Control Approaches to Air Pollution

Conventional pollutants regulations — US

National Ambient Air Quality Standards (NAAQS)

- For each criteria pollutant, the EPA sets a **primary standard** (health, including sensitive populations) and a **secondary standard** (welfare: visibility, vegetation, buildings).
- Standards are expressed as **concentrations** averaged over a specified period (annual and/or short-term, e.g. 3-hour, 8-hour, 24-hour).
- Compliance with short-term averages usually requires the allowable concentration to be exceeded no more than **once per year**.
- Control costs can be quite sensitive to the level of these **short-term averages**.

By statute, the primary standard must be set with an **adequate margin of safety, without regard to cost**, a fundamental departure from the efficiency rule ($MB = MC$).

Conventional pollutants regulations — Europe

EU Ambient Air Quality Directives

- The EU regulates the same criteria pollutants through two directives: 2008/50/EC (PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, lead, benzene) and 2004/107/EC (arsenic, cadmium, mercury, nickel, PAHs).
- Current annual limit values: PM_{2.5} = 25 µg/m³, PM₁₀ = 40 µg/m³, NO₂ = 40 µg/m³.
- A revised directive (2024) will tighten PM_{2.5} to 10 µg/m³ by 2030, closer to the WHO guideline (5 µg/m³).
- Member States must designate air quality zones, monitor concentrations, and prepare Air Quality Plans when limits are exceeded, analogous to US SIPs.
- The European Environment Agency (EEA) coordinates data collection and publishes annual Air Quality in Europe reports.
- Key difference from the US: the EU framework includes exposure reduction targets (national average PM_{2.5} must fall by a specified percentage), not just absolute limits.

Conventional pollutants regulations — Australia

National Environment Protection Measure (NEPM)

- Australia regulates air quality through the **National Environment Protection (Ambient Air Quality) Measure** (AAQ NEPM), set by the National Environment Protection Council.
- Standards exist for $PM_{2.5}$, PM_{10} , O_3 , SO_2 , NO_2 , CO and lead, similar pollutants to the US NAAQS.
- Standards are generally **comparable** to US levels but reflect Australian conditions (e.g. wildfire and dust episodes).
- State/territory EPAs are responsible for **monitoring and enforcement**.

Key difference from the US: Australia does not have a formal **nonattainment** designation system with escalating legal consequences.

Conventional pollutants regulations — comparison

Primary/limit standards for key pollutants

Pollutant	Avg. time	US	EU	Australia	WHO
PM _{2.5}	Annual	9	25 → 10 ^a	7	5
PM _{2.5}	24-hour	35	—	20	15
PM ₁₀	24-hour	150	50	50	45
PM ₁₀	Annual	—	40	20	15
O ₃	8-hour	137	120	128 ^b	100
SO ₂	1-hour	197	350	524	—
NO ₂	Annual	100	40	56	10
CO	8-hour	10.3 mg	10 mg	10.3 mg	—
Lead	Various	0.15	0.5	0.50	—

All values in $\mu\text{g}/\text{m}^3$ unless noted (CO in mg/m^3 ; Lead in $\mu\text{g}/\text{m}^3$). Averaging time: the period over which concentration is measured and averaged, annual (long-term/chronic exposure), 24-hour (short-term), 8-hour (sub-daily), 1-hour (peak/acute). US values converted from ppm/ppb at 25°C, 101.3 kPa. ^aEU revised directive (2024/2881): 10 $\mu\text{g}/\text{m}^3$ by 2030. ^bAU O₃: original NEPM used 4-hour avg; 8-hour standard added in 2021 variation. Lead: US rolling 3-month, EU/AU annual. AU values reflect 2021 NEPM variation (effective 2025).

Sources: US EPA NAAQS (2024); EU Directive 2008/50/EC & 2024/2881; Australia NEPM AAQ (2021 variation); WHO AQG (2021).

PM_{2.5} standards across jurisdictions

A global pattern of progressive tightening

Jurisdiction	Annual standard	History
US (NAAQS)	9 $\mu\text{g}/\text{m}^3$	15 (1997) → 12 (2013) → 9 (2024)
Australia	8 $\mu\text{g}/\text{m}^3$	advisory (2003) → 8 (2016); goal 7 (2025)
EU	25 $\mu\text{g}/\text{m}^3$	25 (2008) → 10 by 2030 (Dir. 2024/2881)
WHO guideline	5 $\mu\text{g}/\text{m}^3$	10 (2005) → 5 (2021)

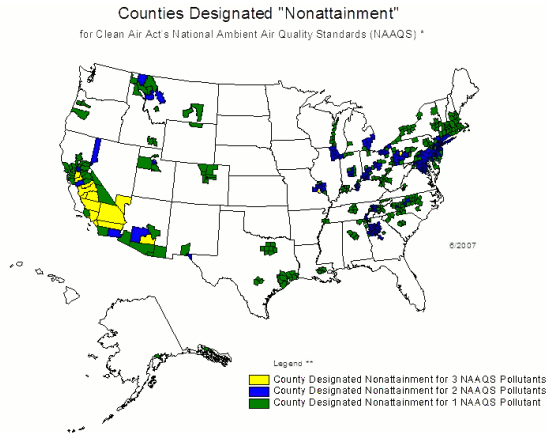
Every jurisdiction is tightening as evidence mounts that there is **no safe threshold** for PM_{2.5}. Australia's standard (8 $\mu\text{g}/\text{m}^3$) is currently **stricter** than the US (9) and the EU (25). Yet no jurisdiction meets the 2021 WHO guideline of 5 $\mu\text{g}/\text{m}^3$. The EU's high limit reflects **political compromise**: coal-dependent members (Poland, Czech Republic) would face immediate nonattainment under stricter standards.

Implementation: SIPs and nonattainment

Who enforces the standards?

- The EPA sets the **ambient standard**; **states** enforce it via **State Implementation Plans (SIPs)**, dividing territory into air-quality control regions.
- Regions that **fail** to meet the standard are designated **nonattainment** areas and face stricter controls.
- Nonattainment areas fall into categories of severity, the more severe, the more stringent the control requirements.
- The EPA can deny **federal highway funds** and other grants to states that fail to provide adequate plans.

Nonattainment areas — all NAAQS pollutants (2007)



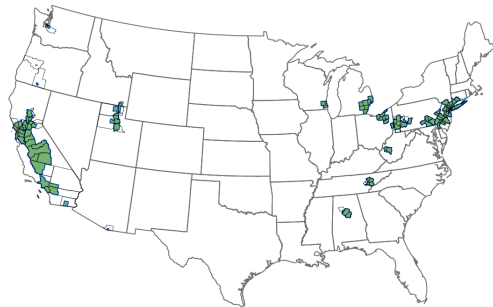
Counties colour-coded by number of NAAQS pollutants in nonattainment: green = 1, blue = 2, yellow = 3.

California's Central Valley and the Northeast corridor are the most affected regions.

Source: [US EPA Green Book](#) (public domain), June 2007.

Nonattainment areas — PM_{2.5} (2010 vs 2018)

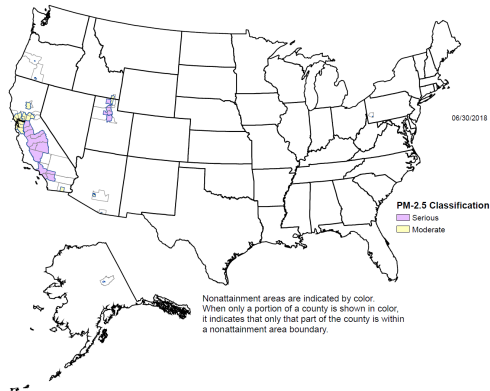
PM-2.5 Nonattainment Areas (2006 Standard)



Nonattainment areas are indicated by color.
When only a portion of a county is shown in color,
it indicates that only that part of the county is within
a nonattainment area boundary.

6/2010

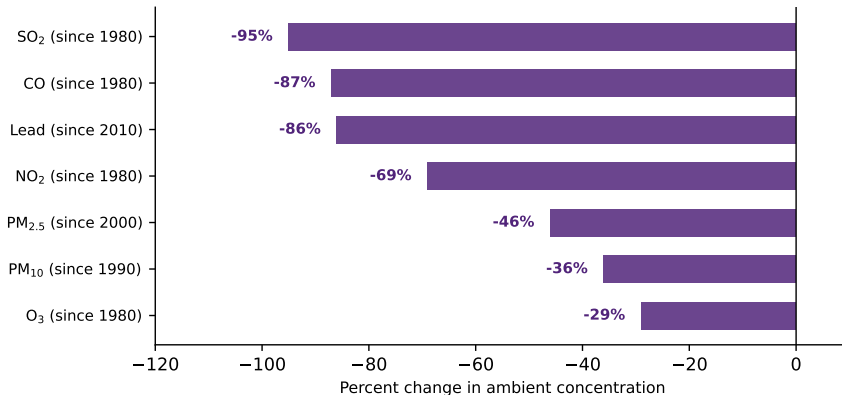
PM-2.5 Nonattainment Areas (2006 Standard)



By 2018, most eastern counties had returned to attainment, only California's Central Valley remained in serious nonattainment. This reflects the success of SO₂ and NO_x controls in reducing secondary PM_{2.5} formation.

Source: [US EPA Green Book](#) (public domain). Left: June 2010. Right: June 2018.

US air quality trends — by pollutant



All criteria pollutants have declined, but the gains are **uneven**: SO₂ (−95%) and CO (−87%) improved most, while ozone (−29%) and PM_{2.5} (−46%), the pollutants most harmful to health, have proved harder to reduce. Part of the improvement reflects **structural change** (shift from manufacturing to services), not just policy. Many US counties still fail NAAQS for O₃ and PM_{2.5}.

Source: EPA, “Air Quality — National Summary” (2025). GDP: BEA (1970–2024).

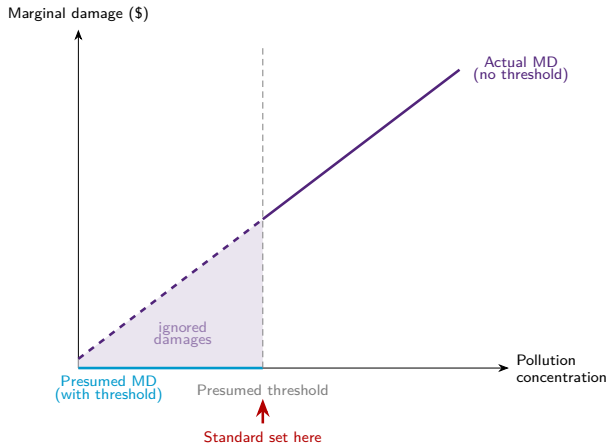
Is the command-and-control approach efficient? (1/3)

How is the standard level set?

- The Clean Air Act prohibits the balancing of costs and benefits in setting ambient standards.
- Instead, the primary standard is set using a health threshold: “a margin of safety sufficiently high that no adverse health effects would be suffered by any member of the population.”
- This presumes a safe concentration below which there is no harm. If true, the marginal damage function would be zero below the threshold and positive above it.

For $PM_{2.5}$ and ozone, the evidence suggests no safe threshold, the dose–response curve starts at zero. This undermines the entire logical basis for the threshold approach.

Is the command-and-control approach efficient? (2/3)



- The Clean Air Act assumes a **safe threshold** below which pollution causes no harm.
- If true, $MD = 0$ below the threshold (solid cyan line).
- But for $PM_{2.5}$ and ozone, evidence shows **no safe threshold**, damage starts at zero (dashed purple line).
- Setting the standard at the presumed threshold **ignores** all damages in the shaded region.
- Without a threshold, efficiency requires **balancing** $MB = MC$, which the statute **forbids**.

Is the command-and-control approach efficient? (3/3)

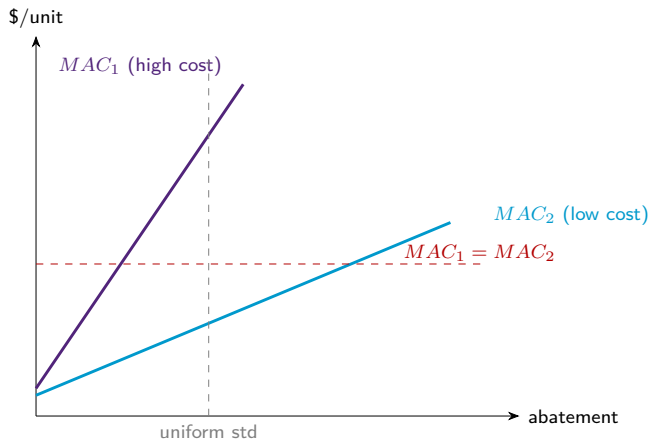
1. **The level:** because costs are excluded, there is no guarantee the standard is at $MB = MC$. EPA benefit–cost reviews show aggregate benefits **exceed** costs, but we cannot confirm **efficiency** at the margin.
2. **Uniformity:** the same standard everywhere ignores differences in **population exposure**, **ecosystem sensitivity**, and **compliance cost** across regions. A standard that is efficient for New York may be too strict for rural Montana.
3. **Timing:** concentrations fluctuate, but the Clean Air Act rules out **intermittent controls**, constant controls raise compliance costs.
4. **Indoor vs outdoor:** the average person spends only $\sim 10\%$ of time outdoors (Klepeis et al. 2001), yet indoor air quality is **not regulated** by the Clean Air Act.

Is the command-and-control approach cost-effective? (1/2)

Usually not

- The cost-effectiveness of CAC depends on local circumstances: prevailing meteorology, the spatial configuration of sources, stack heights, and the degree to which costs vary across sources.
- Simulation models have compared the cost of meeting air-quality targets under CAC (uniform emission standards) to the **least-cost** allocation (equimarginal).
- The ratio of CAC cost to least cost has been estimated at anywhere from **1.1×** to **22×** (Tietenberg & Lewis, Table 15.2).
- CAC **approaches** cost-effectiveness only when a **high degree of control** is needed, such that all sources must abate as much as is economically feasible.

Is the command-and-control approach cost-effective? (2/2)



A **uniform standard** forces both sources to the same abatement level. The equimarginal rule shifts effort to the **low-cost** source, same total abatement, **lower cost**. The wider the spread in MAC, the bigger the waste from uniformity.

The command-and-control approach

The bottom line on command-and-control for air

- CAC is only **close to cost-effective** if a high degree of control is needed such that all sources abate as much as is economically feasible.
- While **inefficient**, CAC policies have resulted in **much better air quality** in developed countries.
- US ambient concentrations of criteria pollutants have fallen **29–95%** since 1980, even as real GDP grew **338%**.
- Developing countries, however, need to find **cost-effective** ways to improve air quality, budgets are tight and the health burden is enormous.
- This motivates the search for **innovative** (market-based) approaches to reduce the cost of pollution control.

Market-Based Approaches to Air Pollution

Market-based approaches — overview

Emissions charging versus emissions trading

- Many examples in the US and around the world of successful market-based instruments for air pollution.
- Markets have been especially successful for air pollution because:
 - stationary sources have measurable point emissions (CEMS);
 - large cost heterogeneity across sources creates gains from trade;
 - uniform-mixing assumptions hold reasonably for some pollutants (SO₂ for acid rain).
- Two main designs: emission charges (taxes) and emission trading (cap-and-trade, credit trading).

Emission charges for air pollution

Two types of charges

- An **efficiency charge** is set to achieve the efficient outcome by forcing the polluter to compensate completely for all damage ($t = MD$ at the optimum).
- A **cost-effective charge** is designed to achieve a given ambient standard at the **lowest possible cost** (equimarginal).
- In practice, charges must be set by an **administrative process**, unlike trading, where the market discovers the price.

Practical examples

- **France**: SO₂ charge on ~1,400 firms; revenues returned as subsidies for abatement equipment. Charge too **low** to change behaviour, a revenue instrument, not an incentive instrument (Harrington, Morgenstern & Sterner 2004).
- **Japan**: SO₂ charge funds compensation for pollution-related health injuries (1973 law). Revenue needs rising, emissions falling (Imura & Schreurs 2005).

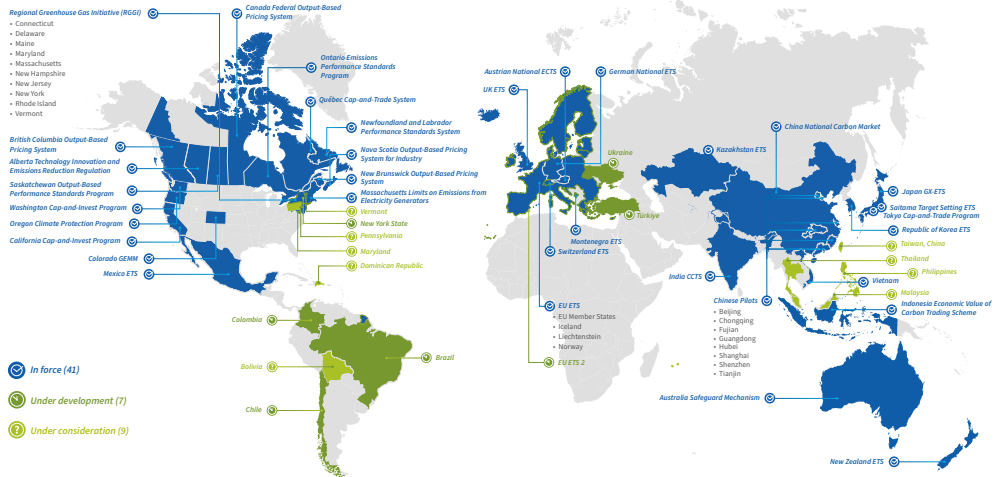
Emission trading mechanisms

Four early forms of trading in the US

1. **Offset**: a new source in a nonattainment area can enter only if it “offsets” its emissions by purchasing reductions from an existing source.
2. **Bubble**: treats an entire plant (or multiple sources) as a single “bubble”, the firm meets an overall cap and allocates reductions internally.
3. **Netting**: when modifying a source, the firm can avoid NSR by netting increases against decreases elsewhere in the same plant.
4. **Banking**: firms can store unused **emission reduction credits** (ERCs) for future use or sale.

These mechanisms added flexibility within the existing CAC framework but did not replace it. The real shift came with the **cap-and-trade** design of the Acid Rain Program: a firm, declining aggregate cap with freely tradeable allowances.

Emissions trading systems worldwide (2026)



41 ETS currently in force, 7 under development, 9 under consideration, covering 26% of global GHG emissions (~14 GtCO₂e) and 63% of global GDP.

Source: ICAP Status Report 2026. International Carbon Action Partnership (icapcarbonaction.com).

The SO₂ Acid Rain Program — background

The problem

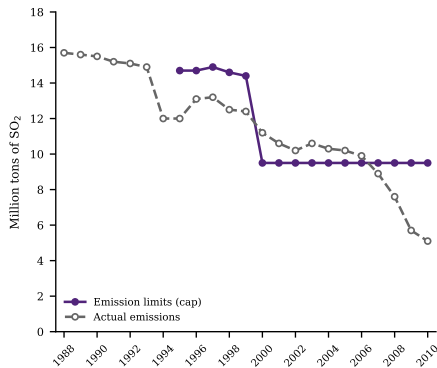
- **Pollutant:** sulfur dioxide (SO₂) and nitrogen oxides (NO_x).
- **Sources:** coal-fired electricity generating plants, primarily in the US Northeast and Midwest.
- **Damage:** acid rain causes acidification of lakes and streams, damages trees at high elevations, accelerates decay of buildings and monuments, and harms human health via secondary PM_{2.5}.
- **Policy:** Clean Air Act Amendments of 1990 (Title IV).
- **Goal:** reduce total annual US SO₂ emissions by 10 million tons relative to 1980 (≈50% reduction).



Source: Wikimedia Commons, CC BY 2.0. Dead spruce from acid rain, Whitetop Mountain, Virginia.

The SO₂ Acid Rain Program — design

- Phase 1 (1995–1999): targeted the 263 most polluting plants.
- Phase 2 (2000–): national cap covering ~3,200 plants with a declining cap over time (permits had a “vintage”).
- Allowances were freely allocated based on historical heat input.
- Banking allowed, and heavily used, smoothing the early price.
- Continuous emissions monitoring (CEMS) provided credible monitoring, reporting, and verification (MRV), a key enabler.
- Penalty: \$2,000/ton for excess emissions (adjusted for inflation), *plus* forfeiture of equivalent tonnes the next year.



Source: Schmalensee & Stavins (2013), Figure 1. Data: Ellerman (2003); US EPA (2012).
Note: Emission limits for 1995–1999 = Phase 1 units' cap + Phase 2 units' emissions. Actual emissions for all years = Phase 1 + Phase 2 units.

The SO₂ Acid Rain Program — the auction innovation

Making prices transparent

- Each year the EPA withheld 2.8% of allocated allowances for a sealed-bid auction.
- Allowances went to the highest bidders, each paying their own bid price (not a uniform clearing price).
- Proceeds were refunded to the utilities from whom allowances were withheld, on a proportional basis.
- Private offerings: sellers could also submit minimum acceptable prices; EPA matched remaining bids with the lowest asking prices.

The main advantage: price transparency. Publicly available auction prices facilitated investment decisions and built confidence in the market.

The SO₂ Acid Rain Program — performance

Results (Schmalensee & Stavins 2013)

- Between 1990 and 2004, SO₂ emissions from power plants decreased 36% while electricity generation increased 25%.
- The long-term goal (10 million ton reduction) was achieved by 2006, ahead of schedule.
- Cost savings compared with command-and-control estimated at at least 15% (and perhaps as much as 90%).
- Evidence that the program spurred innovation and diffusion of new abatement approaches.

The SO₂ Acid Rain Program — benefits

- The 10-million-ton target was selected largely on the basis of the “elbow” of the abatement cost curve.
- The main intended purpose was to reduce the acidification of forest and aquatic ecosystems.
- But these ecological benefits were relatively small due to long time frames needed to reverse damage.
- More than 95% of the monetised benefits came from human health improvements — reduced PM_{2.5} from lower SO₂, a co-benefit that was not the primary target.

Estimated annual US benefits and costs of the SO₂ Allowance Trading Program

Benefits (bn USD 2000)	
Mortality	50–100
Morbidity	3–7
Recreational visibility	2–3
Residential visibility	2–3
Ecosystem effects	0.5
Total	59–116
Costs	0.5–2.0
Net benefits	58–114

Source: Burtraw et al. (1998); Burtraw (1999); Chestnut & Mills (2005); Banzhaf et al. (2006).

The SO₂ Acid Rain Program — how abatement happened

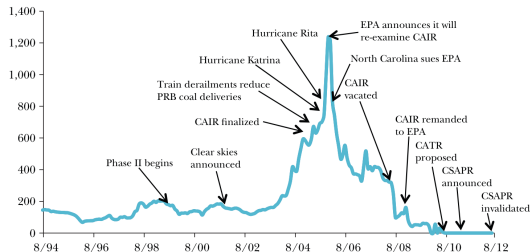
Innovation and discovery

- The programme provided strong incentives for innovation because every tonne reduced was a tonne that could be banked or sold.
- Utilities learned to:
 - burn cost-effective mixtures of different types of coal;
 - build more cost-effective flue gas desulfurisation (scrubber) units;
 - switch to low-sulfur Powder River Basin coal delivered by rail, a cheap option the market discovered.
- Railroad deregulation (Staggers Act 1980) had already lowered rail freight costs, making coal switching attractive, a policy interaction unrelated to the environment.

The SO₂ Acid Rain Program — demise

Court rulings replaced the national market

- The **Clean Air Interstate Rule** (CAIR, 2005) and then the **Cross-State Air Pollution Rule** (CSAPR, 2011) effectively **replaced** the national SO₂ market with regional rules.
- Allowance prices **collapsed** after 2010 as the market became irrelevant.
- The collapse was driven by **legal and regulatory** changes, not by a failure of the market mechanism itself.



SO₂ allowance prices and the regulatory environment, 1994–2012 (1995 USD/ton).

Source: Data on spot prices compiled by Power & Energy Analytic Resources (PEAR) Inc. from Cantor Fitzgerald until September 11, 2001, and from ICAP United thereafter. Notes: CAIR is “Clean Air Interstate Rule.” CATR is “Clean Air Transport Rule.” CSAPR is “Cross-State Air Pollution Rule.” Schmalensee & Stavins (2013), Figure 2.

Lessons from the “grand policy experiment”

Schmalensee & Stavins (2013) — four lessons

1. Much is often learned over time about a policy's benefits and costs $\Rightarrow$ **flexibility** is important.
2. **Unintended consequences** of policies are almost inevitable (rail deregulation, co-benefits, cap rigidity).
3. In most cases, **politics trumps science and economics** (the programme was ended by court order, not by its failure).
4. **Market-based policies** have great cost and feasibility advantages, but like any public policy, the government can change, repeal, or render them irrelevant.

Co-benefits and co-costs

When one policy hits multiple targets

- **Co-benefits:** the SO₂ allowance program targeted **acid rain** (ecological), but >95% of monetised benefits came from **human health** improvements (lower PM_{2.5}).
- **Co-costs:** the 1990 Clean Air Amendments required oxygenated fuels in nonattainment areas. Most chose **MTBE** (cheapest), which turned out to be a major **groundwater** contaminant, a co-cost that was not foreseen.
- Sometimes co-benefits or co-costs may not be recognised when the **ex ante** analysis is conducted. Where they *can* be foreseen, **multiple objectives** can be achieved with a single policy at a lower cost than tackling each issue separately.
- **Climate policy** is the leading example: reducing fossil fuels cuts CO₂ *and* local air pollutants. Ignoring co-benefits **understates** the net benefit of action.

Markets lower the capacity a regulator needs

	Command-and-control	Emissions market
Regulator's task	prescribe equipment, firm by firm	set total cap; firms choose how
Compliance	binary (pass/fail); severe penalties	proportional fines, automatic
Monitoring	manual inspection by staff	continuous (CEMS), automated
Information	regulator must know each firm's costs	market discovers the price
Innovation	no incentive beyond compliance	every tonne reduced is bankable

Markets **economise on state capacity**: the regulator needs to monitor emissions and enforce the cap, but does not need to know costs or prescribe technology. This matters most where state capacity is weakest.

Markets worked where CAC had failed — Surat

- Designed with the Gujarat Pollution Control Board and evaluated as a **randomised trial** across industrial plants (Greenstone et al. 2025).
- Results:
 - **99%** permit compliance (vs 66% under CAC).
 - **–20–30%** pollution emissions relative to control plants.
 - **–11%** abatement costs, holding emissions constant.
 - **$\geq 25\times$** benefit–cost ratio from mortality reduction.
- The market **simplified** the regulator's job: read CEMS data and verify permits, no need to inspect technology or prescribe abatement strategies plant by plant.



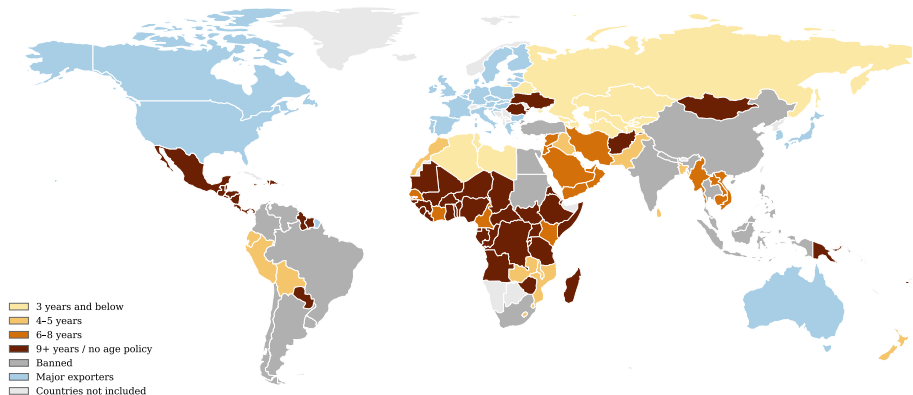
Industrial pollution near Mumbai. Photo: Sumaira Abdulali, Wikimedia Commons (CC BY-SA 4.0).

CEMS: Continuous Emissions Monitoring Systems, sensors on each smokestack that report pollution in real time.

Transboundary pollution spillovers

- $1 \mu\text{g}/\text{m}^3$ more $\text{PM}_{2.5}$ in China $\Rightarrow +0.6\%$ mortality in downwind South Korea (Heo, Ito & Kotamarthi 2025).
- China's "war on pollution" (2014–) cut South Korean $\text{PM}_{2.5}$ by $9.6 \mu\text{g}/\text{m}^3$: ~ 300 fewer deaths per million per year, worth $\sim \text{USD } 2.6$ billion/year.
- But China did **not target** the regions where total (cross-border) benefits were largest.
- **Implication:** air quality policy has large **positive externalities** across borders, international coordination (or side payments) could improve outcomes for both countries.

Rich countries export their vehicle problem



The EU, Japan, and the US exported **14 million** used vehicles in 2015–2018; **70%** went to low-income countries. Many importers set *no* vehicle age limit at all — older, more polluting vehicles concentrate in countries with the weakest air-quality standards.

Source: UNEP (2020), *Used Vehicles and the Environment*. Used light-duty vehicle import age limits, July 2020.

Congestion pricing: a market-based success for urban air

Pricing road access to reduce emissions

City	Year	PM effect
London	2003	PM ₁₀ -12–22%; congestion -26%
Milan	2008/12	suspension $\Rightarrow$ PM ₁₀ +16%, PM _{2.5} +21%
Stockholm	2006	pollution -5–15%; child asthma $\downarrow$
Singapore	1998	electronic road pricing; charges vary by vehicle

Congestion pricing is a **location-specific Pigouvian tax**: it prices the externality (congestion, emissions) at the point of use. Milan's natural experiment (suspension during a strike) confirmed the causal effect. Few developing-country cities have adopted full congestion pricing, though many use licence-plate rationing (Beijing, Bogotá, Santiago, Mexico City) or time-of-day toll highways (Santiago) instead.

Appendix

Key environmental agencies worldwide (1/4)

National / federal

- **US EPA**: Environmental Protection Agency — sets and enforces NAAQS, administers the Clean Air Act and Clean Water Act
- **EEA**: European Environment Agency — data and assessments supporting EU environmental policy
- **DCCEEW**: Dept of Climate Change, Energy, the Environment and Water (AU) — administers EPBC Act, National Pollutant Inventory
- **EA**: Environment Agency (UK) — regulates industrial emissions, water quality, flood risk
- **MOE**: Ministry of the Environment (Japan) — air/water quality standards, Minamata Convention host
- **MEE**: Ministry of Ecology and Environment (China) — national emission standards, real-time monitoring network

Key environmental agencies worldwide (2/4)

International / multilateral

- **UNEP**: UN Environment Programme — coordinates global environmental action, hosts multilateral environmental agreements
- **WHO**: World Health Organization — publishes ambient air quality guidelines (updated 2021)
- **IPCC**: Intergovernmental Panel on Climate Change — assessment reports on climate science and mitigation
- **OECD**: Organisation for Economic Co-operation and Development — Environmental Policy Stringency Index, green growth indicators
- **World Bank**: environmental economics research, Worldwide Governance Indicators, pollution data
- **IUCN**: International Union for Conservation of Nature — Red List, protected area standards

Key environmental agencies worldwide (3/4)

Australian state and territory EPAs

- DES — Queensland Department of Environment and Science
- EPA NSW — New South Wales Environment Protection Authority
- EPA VIC — Environment Protection Authority Victoria
- EPA SA — South Australian Environment Protection Authority
- DWER — Western Australia Dept of Water and Environmental Regulation
- NT EPA — Northern Territory Environment Protection Authority
- EPA TAS — Tasmanian Environment Protection Authority
- ACT EPA — ACT Environment Protection Authority

Key environmental agencies worldwide (4/4)

Other notable agencies

- **CPCB** (India) — Central Pollution Control Board
- **ECCC** (Canada) — Environment and Climate Change Canada
- **IBAMA** (Brazil) — Brazilian Institute of Environment and Renewable Natural Resources
- **UBA** (Germany) — Federal Environment Agency
- **MoEFCC** (India) — Ministry of Environment, Forest and Climate Change
- **MTE** (France) — Ministry of Ecological Transition
- **MOE** (South Korea) — Ministry of Environment

Data sources for air pollution research

Air pollution data

- WHO Global Health Observatory — ambient air quality database ($\text{PM}_{2.5}$, PM_{10} , NO_2) for 6 000+ cities
- ACAG / van Donkelaar — satellite-derived global $\text{PM}_{2.5}$ grids (1998–present, ~ 1 km resolution)
- Our World in Data — country-level $\text{PM}_{2.5}$ exposure, deaths, and trends (sourced from IHME/GBD)
- US EPA AQS — Air Quality System: hourly monitor data across the US
- WAQI / AQICN — real-time Air Quality Index from 30 000+ stations worldwide
- OECD Air Quality and Health — mortality and welfare cost estimates from air pollution
- EDGAR — Emissions Database for Global Atmospheric Research (CO_2 , CH_4 , N_2O , SO_2 , NO_x)
- State of Global Air — IHME/HEI: country profiles on air quality and health burden

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